

S M Jishanul Islam

Software Engineer, Apurba Technologies Ltd., Dhaka, Bangladesh

jishanlion@gmail.com — +880 1759338652 — [Linkedin](#) — [Portfolio](#) — [GitHub](#) — [Google Scholar](#)

RESEARCH INTERESTS

Machine Learning - Multimodal Deep Learning - Multimodal Representation.

EDUCATION

United International University, Dhaka, Bangladesh

Bachelor of Science in Computer Science and Engineering (**Major**: Data Science)

Thesis Title: AVLoS: Audio-Visual Long Text Scene Summarization

Thesis Supervisor: Prof. [Dr. Swakkhar Shatabda](#)

2020 — 2024

CGPA: 3.77/4.00

Major CGPA: 4.00/4.00

EXPERIENCE

Apurba Technologies Ltd.

Software Engineer

Dhaka, Bangladesh

December 2024 — Present

- Built scalable RESTful APIs and deployed them to local and cloud servers.
- Optimized query and API performance, reducing retrieval and data creation times by 10% and increasing transactions per second by implementing indexing strategies and caching mechanisms.
- Collaborated with companies and government agencies to implement, research, and complete project deliverables 2-3 weeks before the deadline.

United International University

Lecturer (contractual)

Dhaka, Bangladesh

June 2024 — October 2024

- Taught core CS concepts to students.
- Took class tests and term exams to monitor students' performance and provide informative feedback, improving average student performance by 7%.
- Collaborated with the Dept. of CSE to audit students' marks for BAETE/IEB accreditation.
- **Courses conducted:** Object-Oriented Programming Lab, Data Structures and Algorithms 1 Lab, and Database Management Systems Theory.

United International University

Undergraduate Assistant

Dhaka, Bangladesh

September 2023 — May 2024

- Managed to facilitate course content and materials with the lab faculty, checked and evaluated assessments, and judged lab projects.
- Created the first course materials for the country's first-ever undergraduate degree in Data Science.
- **Courses supervised:** Programming for Data Science, and Object-Oriented Programming for Data Science.

PUBLICATIONS

Journal papers

Under Review

- **S M Jishanul Islam**, Sahid Hossain Mustakim, Musfirat Hossain, Mysun Mashira, Nur Islam Shourav, MD. Rayhan Ahmed, Salekul Islam, Swakkhar Shatabda, and A.K.M. Muzahidul Islam. 2024. An audio video-based multi-modal fusion approach for speech emotion recognition. Knowledge-Based Systems (Elsevier).

Conference papers

Published

- Sadia Ahmmed, Taimur Rahman, **S M Jishanul Islam**, Al-Momen Reyad, Sonjoy Dey, James Anthony Purification, and Md. Dewan Farid. 2024. E-MedViTR - Enhanced Vision Transformers with Registers for Biomedical Image Classification. 6th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT 2024) [Scopus Indexed]. DOI: [10.1109/ICEEICT62016.2024.10534573](https://doi.org/10.1109/ICEEICT62016.2024.10534573)

Workshop papers

Accepted

- **S M Jishanul Islam**, Sahid Hossain Mustakim, Sadia Ahmmed, Md. Faiyaz Abdullah Sayeedi, Swapnil Khandoker, Syed Tasdid Azam Dhruho, and Nahid Hossain. 2024. MIMIC: Multimodal Islamophobic Meme Identification and Classification. 3rd Muslims in ML Workshop - Neural Information Processing Systems 2024 (NeurIPS 2024). DOI: [10.48550/arXiv.2412.00681](https://doi.org/10.48550/arXiv.2412.00681)

RESEARCH EXPERIENCE

An audio video-based multi-modal fusion approach for speech emotion recognition.

- **Co-supervised by:** [Md. Rayhan Ahmed](#), [Dr. Salekul Islam](#), [Dr. Swakkhar Shatabda](#), and [Dr. A.K.M Muzahidul Islam](#).
- Under Review at Knowledge-Based Systems (Elsevier) [[codebase](#)].

- We present an approach to classify human emotions by fusing audio and visual inputs. Our approach sets new state-of-the-art results while keeping the architecture very simple. Additionally, we present a new frame filtering strategy to overcome the problem of spatiotemporal redundancy.

AVLoS: Audio-Visual Long Text Scene Summarization

- **Supervised by:** [Dr. Swakkhar Shatabda](#).
- Thesis Work. In Progress.
- We are experimenting on fine-grained long text scene summarization from videos using video and audio inputs. Most approaches use queries to guide text generation, we are experimenting to generate text without these queries.

MIMIC: Multimodal Islamophobic Meme Identification and Classification

- **Supervised by:** [Nahid Hossain](#).
- Accepted at MusIML Workshop - NeurIPS 2024. Future work for version 2. [[arXiv version](#)] [[OpenReview](#)] [[codebase](#)].
- Anti-Muslim hate speech has emerged within memes, characterized by context-dependent and rhetorical messages using text and images that seemingly mimic humor but convey Islamophobic sentiments. This work presents a novel dataset and proposes a classifier based on the Vision-and-Language Transformer (ViLT) specifically tailored to identify anti-Muslim hate within memes by integrating both visual and textual representations.

BhaShammo: IPA Transcription of Bengali Regional Dialect using Dialect Guided Tokens

- **Co-supervised by:** [Dr. Swakkhar Shatabda](#) and [Dr. Farig Sadeque](#).
- In Progress. [[arXiv version](#)] [[codebase](#)].
- We present an approach to transcribe regional Bengali text to IPA by introducing the Dialect Guided Tokens (DGT) technique on a new dataset spanning six districts of Bangladesh. We provide the model with information on the regional dialect of the input text before generating the IPA transcription. This is the first time this problem has been solved.

E-MedViTR: Enhanced Vision Transformers with Registers for Biomedical Image Classification

- **Supervised by:** [Dr. Dewan Md. Farid](#).
- Published at ICEEICT. [[paper](#)].
- We investigated the effectiveness of the ViT with registers (aka DINO v2) in classifying medical pathology images. Normally, it doesn't perform up to the mark as SOTA models do, but with an extension and data augmentations, it performs relatively close to SOTA models.

PROJECTS

NurtureAid (github.com/sadia-ahmed/Agewell-Frontend)

Built a real-time AI-based cross-platform mobile application that simplifies care through caretakers. Led the app's backend and frontend development as a full-stack engineer.

Tech Stack: React Native, Node.js, Flask, MongoDB, Firebase, LLaMA-Index, PyTorch.

Acknowledgements: Champion of the UIU CSE Project Show, Software Engineering Laboratory; Champion of the Hult Prize OnCampus round in UIU; Selected for the Hult Prize Summit in Boston.

IPBlocks (closed for academic review)

Developed a web3 dApp that helps users protect, license, and commercialize Intellectual Property (IP) through blockchain technology. Led the development of the app's backend, frontend, and blockchain development as a full-stack engineer.

Tech Stack: HTML, CSS, JavaScript, PHP, PostgreSQL, Solidity, Truffle, Ganache.

Acknowledgements: Gold Award (Champion) at the International Blockchain Olympiad 2023.

Quest Aid (github.com/sadia-ahmed/quest-aid)

Developed an AI-driven web application to manage the ECAs of students, clubs, universities, and organizations under the same platform. Managed the app's backend, frontend, and AI development as a full-stack engineer..

Tech Stack: React.js, Spring Boot, Flask, MySQL, Langchain.

iamspecial.com (github.com/S-M-J-I/iamSpecial-dbms-project)

Developed a web app built to tackle the lack of misinformation, accessibility and guidance for people with special needs. Led the app's backend and frontend development as a full-stack engineer.

Tech Stack: HTML, CSS, JavaScript, PHP, MySQL.

Acknowledgements: Champion of the UIU CSE Project Show, Database Management Systems Laboratory.

SKILLS

Programming: C, C++, Java, Python, JavaScript, PHP, SQL, Solidity.

Machine Learning: PyTorch, Tensorflow, Scikit-Learn, Matplotlib, Numpy, Pandas, NLTK, HuggingFace.

Frameworks/Libraries: React.js, Node.js, Spring Boot, FastAPI, Firebase.

Research Skills: L^AT_EX, Data Visualization, Data Interpretation, Methodology Formulation.

DevOps Tools: Git, GitHub, Docker.

Soft Skills: Communication, Leadership, Design Thinking, Public Speaking, Team Player.

ACHIEVEMENTS

Winner, Harvard Health Systems Innovation Lab Hackathon 2025 (Dhaka Hub)

Academic Excellence Scholarship

Gold Award (Champion), International Blockchain Olympiad (IBCOL) 2023

April 2025

2020 — 2024

November 2023

Champion, Bhashamul: Bengali Regional Text to IPA National NLP Datathon	March 2024
Champion, Hult Prize OnCampus Round	March 2024
Representative, Hult Prize Summit in Boston, USA	June 2024
Silver Award(1 st Runner Up), Bangladesh Blockchain Olympiad (BCOLBD) 2023	July 2023
1 st Runner Up, Intra-University Deep Learning Sprint	January 2023
Finalist, ICT Innovation Grant (for NurtureAid)	December 2023
Champion, Software Engineering Lab	September 2023
Champion, Database Management Systems Lab	May 2022
Champion, Advanced Object Oriented Programming Lab	January 2022
The Daily Star Award	2017
Academia High Achievers Award	2017

EXTRA CURRICULARS

UIU Computer Club - served as the General Secretary	2022 — 2024
UIU Entrepreneur Development Forum - served as the Organizing Secretary	2020 — 2022
Triumph of IBCOL 2023 and the Future of Blockchain in Bangladesh by Jamuna TV - interview link .	
Trainer, Hands-on Workshop on Automatic Speech Recognition by Bengali.AI and IUT Computer Society - workshop link .	
Trainer, Hands-on Workshop on building a dApp from scratch by UIU Computer Club - workshop link .	

REFERENCES

Dr. Swakkhar Shatabda

*Professor of Computer Science and Engineering,
Brac University, Dhaka, Bangladesh*
E-mail: swakkhar.shatabda@bracu.ac.bd

Md. Rayhan Ahmed

*Ph.D. Student, University of British Columbia, Canada
Assistant Professor (on study leave) of Computer Science and
Engineering, United International University*
E-mail: md.ahmed@ubc.ca